Unit: Unit 1: Ratio, Rate, Percentage
Topic: Percent

Name: ________________
Date: __________

Homework (Carolina Reaper)

Q1. A painter uses 25% of blue paint on a canvas. If the canvas is 120 square feet, how many square feet of blue paint are used?
(A) 20
(B) 24
(C) 30
(D) 32
(E) 36

Q2. The frequency of a rhythm pattern is increased by 15%. If the original frequency was 80 Hz, what is the new frequency?
(A) 86 Hz
(B) 88 Hz
(C) 90 Hz
(D) 92 Hz
(E) 96 Hz

Q3. A music festival tickets are priced at \$50. If there is a 12% discount for students, how much will a student pay for a ticket?
(A) \$43.60
(B) \$44
(C) \$46
(D) \$48
(E) \$50

Q4. A canvas has a width of w feet and a length of $2w$ feet. If the canvas is reduced by 20% in width and 10% in length, what is the new area of the canvas in terms of w?
(A) $1.6w^2$
(B) $1.7w^2$
(C) $1.8w^2$
(D) $1.9w^2$
(E) $2w^2$

Q5. A painter mixes 25% red paint with 75% blue paint. If the total amount of paint is 200 gallons, how many gallons of red paint are used?
(A) 40
(B) 50
(C) 60
(D) 75
(E) 100

Q6. The size of a musical instrument is increased by 30%. If the original size was x cubic feet, what is the new size in terms of x?
(A) $0.7x$
(B) $1.1x$
(C) $1.3x$
(D) $1.5x$
(E) $1.7x$

Q7. A sculptor uses 40% of clay for a project. If 30% of the clay is wasted, what percentage of the original clay is used for the project?
(A) 24%
(B) 26%
(C) 28%
(D) 30%
(E) 32%

Q8. A rhythm pattern has 480 beats per minute. If the frequency is decreased by 20%, how many beats per minute are in the new rhythm pattern?
(A) 352
(B) 384
(C) 416
(D) 448
(E) 480

Open Ended Questions (FRQ)

Q9. A painter has 12 gallons of paint. 30% of the paint is used for a mural and 20% is used for a sculpture. If the painter adds x gallons of paint, what is the total percentage of paint used for the mural and the sculpture in terms of x? Show all work and express the answer as a percentage.

Q10. A musical instrument has 48 strings. If 25% of the strings are tuned to a higher frequency and 15% are tuned to a lower frequency, what is the total percentage of strings that are not tuned to the original frequency? If x strings are added to the instrument, what is the new total percentage of strings that are not tuned to the original frequency in terms of x? Show all work and express the answer as a percentage.

Chef's Tips

- Tip 1: Ensure students have a solid understanding of percentages and percent change before attempting this test.
- Tip 2: Encourage students to read each question carefully and identify the key concepts and formulas required to solve the problem.
- Tip 3: Emphasize the importance of checking units and showing all work, especially for free-response questions.